

Supplementary Material for CEARF-N: Bounded Out-of-Fit Dynamic Rank Allocation for Sparse Next-Item Recommendation

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This supplement specifies the complete CEARF-N protocol behind the eight-page short paper: expert scoring, source-disjoint OOF construction, continuous dynamic- β optimization, controls, matched-query statistics, and runtime verification. Neither declared-validation nor test labels fit or select a fusion coefficient.

1 Scope and Claim Ledger

Table 1: What the paper does and does not claim.

Claim	Precise scope
Training-only allocation	The scalar prior and query-specific β_q are fit only from targets of source sessions removed before expert fitting. There is no β grid.
Use of validation	A stable hash first declares 5,000 validation queries from each loader’s candidate pool. This subset may audit the frozen system, and an earlier study used validation to freeze the PASGR cell. It does not train, select, or early-stop β .
Dynamic gain	The primary comparison is dynamic versus an OOF-trained global or short/long policy on identical expert ranks, not dynamic versus a weak endpoint. Video has an interval-above-zero gain; Baby and Diginetica intervals span zero.
Assignment signal	One fixed deterministic reassignment of the exact frozen β_q multiset breaks only query-to-beta matching. Video nDCG@20 and Diginetica nDCG@10 have intervals above zero for this negative control; Baby is unresolved. Its SHA-256 seed uses only the experiment seed and ordered query IDs, never targets.
Metadata attribution	PASGR uses a cache documented as E5-small [14] on Video and TF-IDF/SVD teachers on Baby and Diginetica. Comparisons with ID-only models are system-level; matched-teacher Amazon results are labelled separately.

Table 1 – continued

Claim	Precise scope
Expert sensitivity	The NARM [9] swap changes only the neural expert inside the same allocator and is reported as system-level sensitivity, not as metadata-matched architecture attribution.
Generality	Three domains support a bounded empirical finding, not a literature-wide state-of-the-art or universal routing claim.

1.1 Methodological position

The contribution is not a new encoder, RRF [1], or validation split. It is the combination of frozen heterogeneous experts, source-disjoint OOF supervision, target-free inference features, and a verifiable perturbation bound. Out-of-fit meta-learning is rooted in stacked generalization [16]; our contribution is specifically the five-scalar, bounded allocation law with a rank-only expert interface and its frozen-expert protocol.

Table 2. Allocation design space. “OOF-safe” means the allocator never learns from predictions made on an expert’s own fit sources.

Family	Per-query	Frozen experts	OOF-safe	Bounded
Global weighted rank fusion	No	Yes	N/A	N/A
In-fit stacking/router	Yes	Often	No	Usually no
OOF stacking	Yes	Yes	Yes	Usually no
Joint neural MoE	Yes	No	N/A	Usually no
CEARF-N	Yes	Yes	Yes	Yes

2 Question-to-Control Map

This section maps each research question to the control that changes only one axis of the proposed allocation protocol.

Table 3. Each control changes one methodological axis while preserving the declared train-only allocation boundary.

Question	Control	Interpretation
Is fusion complementary?	Memory-only, neural-only, equal mixing	Separates evidence complementarity from allocator complexity.
Does per-query allocation matter?	OOF global, short/long, head/tail, four buckets	Compares one scalar, coarse policies, and continuous β_q .
Is the assignment itself useful?	Fixed reassignment of primary β_q	Preserves the exact beta multiset while breaking query-to-beta matching.
Is the bound arbitrary?	$\Delta \in \{.05, .10, .20\}$; no-prior/no-admission variants	Tests modulation capacity and both shrinkage terms.
Which signals matter?	Single-feature and drop-one gates; 16-feature linear/MLP	Audits context, popularity/tail, and expert diagnostics.
Is the fusion implementation decisive?	$k \in \{10, 20, 60\}$ and normalized CombSUM	Tests the rank constant and a fixed-allocation score-space perturbation.

3 End-to-End Protocol

1. Declare 5,000 validation queries per domain by stable hash before any allocation fit; exclude their source identifiers from OOF construction.
2. From the remaining training sources, remove up to 5,000 complete sessions: 1,000 for memory-profile locking and 4,000 for gate calibration.
3. Fit CEARF and PASGR on the inner sessions, then predict the two disjoint OOF subsets. Profile selection uses only the first subset.
4. On actionable gate queries, learn the continuous β_{OOF} from reciprocal-rank pairs, then fit the bounded three-feature residual around that frozen prior.
5. Freeze the split fingerprints, expert identities, feature statistics, scalar/gate states, and protocol flags before any declared-validation or test prediction.
6. Within the allocation experiment, use declared validation only for frozen-policy diagnostics. It cannot change β_{OOF} , the gate, Δ , or the primary feature set.
7. Refit experts/indexes on the complete training data permitted for test, retain the frozen allocator, rank the full catalogue, and save per-query ranks for paired analysis.

4 Task, Data Construction, and Evaluation

A query is a prefix $q = (i_1, \dots, i_L)$ with one held-out next item y_q . The catalogue is $\mathcal{I}$, and every reported ranking is over the full catalogue. For Amazon [7],

histories are ordered and split leave-last-out; items already present in the query are masked. For Diginetica, the public session split [2] is retained and repeat consumption is allowed. Candidate generation may operate at width 120 internally, but metrics inspect the final top 6, 10, or 20 positions. No sampled test negatives are used.

Each loader exposes a validation-candidate mapping before this experiment. The study declares exactly 5,000 candidates per domain by stable hash before any allocation fitting. On Diginetica, those 5,000 target events and incoming transitions are removed from the tuning sessions; unselected candidate target events remain ordinary training events. On Amazon, validation targets are external to training, while source histories corresponding to unselected candidates remain OOF-eligible. No unselected candidate is evaluated as part of the declared validation subset.

Table 4. Audited validation boundary. “Candidate–OOF” counts source overlap with the loader’s full candidate mapping; source histories outside the declared 5,000 remain OOF-eligible. Declared–OOF is zero.

Domain	Candidate pool	Declared	Removed	Declared–OOF	Candidate–OOF
Video Games	94,762	5,000	0	0	5,000
Baby Products	150,777	5,000	0	0	5,000
Diginetica	133,724	5,000	5,000	0	5,000

Table 5. Dataset and protocol statistics.

Domain	Items	Train seq.	Validation	Test Seen-item policy
Video Games	25,613	94,762	5,000	94,762 mask
Baby Products	36,014	150,777	5,000	150,777 mask
Diginetica	43,098	186,670	5,000	60,858 retain

For cutoff K , Recall is $|\{q : r_q \leq K\}|/|\mathcal{Q}|$ and nDCG is $|\mathcal{Q}|^{-1} \sum_q \mathbf{1}[r_q \leq K]/\log_2(r_q + 1)$. The allocation objective reported in the paper is $U = .5 \text{ Recall@6} + .5 \text{ Recall@20}$; Recall@10 and all nDCG cutoffs are reported in Tables 9–11.

5 Complete CEARF Memory

5.1 Transition evidence

Only the eight most recent context items are inspected. For context item s_a at reverse-recency age a and candidate x ,

$$T_q(x) = \sum_a e^{-.35a} \frac{\log(1 + c(s_a, x))}{\sqrt{1 + \text{freq}(x)}}. \quad (1)$$

The directed count c sums inverse transition distance within a four-step window. Each source retains at most 200 outgoing candidates.

5.2 Neighbour-session evidence

Let j be an indexed training session. It receives overlap

$$O(q, j) = \sum_a \text{idf}(i_a) e^{-.20a} \mathbf{1}[i_a \in j], \quad \text{idf}(i) = \log \left(1 + \frac{|\mathcal{S}|}{|\{j : i \in j\}|} \right). \quad (2)$$

The 120 highest-overlap sessions vote for candidates. The vote is normalized by $\sqrt{|\text{unique}(j)|}$. An item after the most recent matched position receives direction weight 1; an earlier item receives .25; distance from the match contributes an additional inverse-square-root factor.

5.3 Popularity, profiles, and agreement

Global frequency supplies the third top-120 list and pads a short output. The declared memory family is:

Profile	Transition Session Popularity		
transition	1.00	0	0
session	0	1.00	0
balanced	.50	.40	.10
transition-session	.60	.40	0
session-transition	.35	.60	.05
short-safe	.65	.20	.15

Ranks are combined as $\sum_e w_e / (20 + r_e(x))$. If at least two active memory lists retrieve x , its score is multiplied by $1 + .12(v_x - 1)$, where v_x is its support count. Short ($L \leq 2$) and longer contexts select profiles independently on the 1,000-query training-only profile subset. The selected profile is session-only in all three domains and both regimes; transition and popularity therefore remain evaluated alternatives/fallbacks rather than hidden contributors to the primary result.

6 PASGR Neural Expert

PASGR expands to *Prototype-Aligned Semantic Graph Retrieval*. It is a one-layer GRU with 64 hidden units, dropout .2, candidate width 120, and four training epochs in the dynamic- β experiment. For prefix state $\mathbf{h}_q$ and item embedding $\mathbf{e}_i$, exact full-catalogue inference uses

$$s_N(q, i) = \text{norm}(W \text{GRU}(q))^\top \text{norm}(\mathbf{e}_i).$$

Video Games item embeddings use the cached 384-dimensional teacher documented as **E5-small**. Baby Products uses a 128-dimensional TF-IDF/SVD teacher constructed from available titles, categories, features, descriptions, and brands; Diginetica uses a 128-dimensional TF-IDF/SVD matrix of product-name tokens. Teacher identity is resolved from the loaded matrix before evaluation; its shape and SHA-256 prefix are reported directly in the provenance table in Section 10 rather than inferred from a requested label.

Training contrasts the observed target with 32 negatives drawn from semantic prototype neighbours, transition-graph neighbours, and random catalogue items. Cross-entropy on that candidate set is augmented by a recency-weighted semantic alignment loss and, when enabled, multi-positive in-batch retrieval. Graph aggregation mixes adjacent-item representations before GRU training; prototype transport pulls poorly observed items toward semantic centroids. AdamW uses learning rate 10^{-3} and weight decay 10^{-4} . The upstream PASGR family crossed graph weight $\{0, .35\}$, prototype transport on/off, contrastive weight $\{0, .15\}$, and in-batch weight $\{0, .1\}$. All three frozen cells disable prototype transport, so the “prototype-aligned” mechanism is part of the declared PASGR family but not a hidden contributor to the primary cells.

Table 6. Frozen PASGR cells used by every allocation control. The dynamic experiment does not reselect them.

Domain	Graph	Prototype	Contrastive	In-batch
Video Games	.35	off	.15	.10
Baby Products	0	off	.15	0
Diginetica	.35	off	0	.10

The earlier study evaluated the full 16-cell family on validation. This is an upstream expert-selection decision, not evidence that dynamic β used validation. A pure component-attribution study would need to retrain all cells and evaluate neural-only ranks under the new protocol; we do not relabel the older fusion-selected grid as such an ablation.

7 Training-Only Dynamic Allocation

7.1 Rank evidence and hard negatives

For expert $e \in \{M, N\}$ and rank $r_e(x)$,

$$\rho_e(x) = \begin{cases} (k+1)/(k+r_e(x)), & x \text{ appears in the top 120,} \\ 0, & \text{otherwise,} \end{cases} \quad k = 20. \quad (3)$$

Each calibration query uses the 32 non-target union candidates with largest evidence under either expert, or all candidates when fewer than 32 exist; item ID breaks ties deterministically. A query is actionable when its target appears in at least one expert list. Non-actionable queries are retained in coverage statistics but excluded from optimization because changing β cannot retrieve their missing target from the union.

The serving implementation uses $1/(k+r)$ rather than $(k+1)/(k+r)$. The omitted common positive factor gives $S_q = F_q/(k+1)$ and therefore exactly the same ordering. The expert interface is rank-only: each expert enters allocation through its top-120 ordering, while query features are target-free at inference. Any strictly monotone raw-score transformation that preserves an expert's top-120 ordering therefore preserves the OOF training arrays, learned allocator, and final ranking.

7.2 Global prior and bounded query residual

The global model learns one unconstrained logit whose sigmoid is β_{OOF} , initialized at .35. The query model is

$$\beta_q = \beta_{\text{OOF}} + \Delta_{\text{eff}} \tanh(\mathbf{w}^\top \tilde{\mathbf{z}}_q + b), \quad \Delta_{\text{eff}} = \min\{.10, \beta_{\text{OOF}}, 1 - \beta_{\text{OOF}}\}. \quad (4)$$

The primary feature vector contains exactly:

1. $\log(1+L)$;
2. $\log(1+\text{freq}(i_L))$; and
3. $\mathbf{1}[i_L \notin \mathcal{I}_{\text{head}}]$.

Features are computable before recommendation and contain neither target identity nor a target-hit diagnostic. Mean and standard deviation are fit on actionable OOF training rows and frozen.

For target y_q and negative x , the fused evidence is $F_q(x) = (1 - \beta_q)\rho_M(x) + \beta_q\rho_N(x)$. With temperature $\tau = .10$,

$$\mathcal{L}_{\text{rank}} = \frac{1}{|Q|} \sum_q \frac{1}{|\mathcal{H}_q|} \sum_{x \in \mathcal{H}_q} \text{softplus}\left(\frac{F_q(x) - F_q(y_q)}{\tau}\right), \quad (5)$$

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + 10^{-3} \overline{\beta_q} + 10^{-3} (\overline{\beta_q} - \beta_{\text{OOF}})^2. \quad (6)$$

The first regularizer is a small neural-admission cost; the second shrinks query modulation toward the global policy. AdamW uses learning rate .02 for the 80-epoch scalar fit and 10^{-3} for the 50-epoch gate, batch size at most 512, weight decay 10^{-4} for the gate, and deterministic seed order. The scalar stage minimizes $\mathcal{L}_{\text{rank}} + 10^{-3}\beta$ with $\beta_q \equiv \sigma(a)$; only the quadratic residual-prior term is absent. It is then frozen while the second stage optimizes only $(\mathbf{w}, b)$ under the full objective. For a target-negative pair, define the signed expert-margin advantage

$$A_{qx} = [\rho_N(y_q) - \rho_N(x)] - [\rho_M(y_q) - \rho_M(x)].$$

The derivative of the fused target margin with respect to β_q is A_{qx} : the gate learns from context when neural evidence has the larger target-versus-negative margin.

Table 7. Training-only calibration status, mean (SD) over 3 matched seeds.

Domain	OOF rows	Actionable	Share	β_{OOF}
Video Games	4000 (0.0)	1666 (25.4)	.4165 (.0063)	.57748 (.00705)
Baby Products	4000 (0.0)	845 (13.1)	.2113 (.0033)	.54335 (.00923)
Diginetica	4000 (0.0)	3298 (8.2)	.8245 (.0020)	.41848 (.00284)

7.3 Why the residual is bounded

The same bounded three-feature linear form, feature set, and hyperparameters are primary across all domains and seeds; fitted coefficients vary by replicate. The direct 16-feature MLP is a capacity stress test. Its gains are small and seed-dependent; on Diginetica, three of four MLP variants are lower than the primary and the remaining no-memory-certainty MLP is higher by only .00025 U . The primary gate treats β_{OOF} as a learned prior and permits only a $\pm .10$ correction. This does not guarantee improvement or bound ranking loss; it limits only per-item fused-score perturbation. The linear inside the tanh has four optimized coefficients. Together with the scalar prior, the allocator has five optimized coefficients; six feature means/standard deviations are frozen preprocessing state, not optimized parameters.

Bounded expert-disagreement certificate. Analytically, finite real logits give $\beta_q \in (0, 1)$ because $\Delta_{\text{eff}} \leq \min(\beta_{\text{OOF}}, 1 - \beta_{\text{OOF}})$ and $|\tanh(\cdot)| < 1$; the implementation-safe guarantee is $[0, 1]$, and all persisted allocations are strictly interior. Moreover, with $F_{\text{OOF}}(x) = (1 - \beta_{\text{OOF}})\rho_M(x) + \beta_{\text{OOF}}\rho_N(x)$,

$$|F_q(x) - F_{\text{OOF}}(x)| = |\beta_q - \beta_{\text{OOF}}| |\rho_N(x) - \rho_M(x)| \leq \Delta_{\text{eff}} \leq .10,$$

since each reciprocal-rank evidence lies in $[0, 1]$. More sharply, let $\delta_q = \beta_q - \beta_{\text{OOF}}$, $D_q(x) = \rho_N(x) - \rho_M(x)$, and $m_{\text{OOF}}(a, b) = F_{\text{OOF}}(a) - F_{\text{OOF}}(b)$. Then

$$F_q(a) - F_q(b) = m_{\text{OOF}}(a, b) + \delta_q \{D_q(a) - D_q(b)\}.$$

The baseline order is certified whenever $|m_{\text{OOF}}(a, b)| > |\delta_q| |D_q(a) - D_q(b)|$, and conservatively whenever $|m_{\text{OOF}}(a, b)| > 2\Delta_{\text{eff}}$. If $D_q(a) = D_q(b)$, the pair is allocation-invariant. For a target-negative pair, $D_q(y_q) - D_q(x) = A_{qx}$ and

$$\frac{\partial \ell_{qx}}{\partial \beta_q} = -\frac{A_{qx}}{\tau} \sigma\left(\frac{F_q(x) - F_q(y_q)}{\tau}\right).$$

Thus the same expert-disagreement term determines the learning signal and the maximum permitted pair intervention. This is a score-margin certificate, not a Recall/nDCG guarantee. Conditional on the declared source split and frozen experts, $\beta_{\text{OOF}}, \mathbf{w}, b$ are functions only of OOF training ranks, OOF training targets, and training-derived feature statistics. Declared-validation and test targets therefore cannot affect the allocator parameters. This construction establishes allocation provenance; it does not by itself guarantee test improvement.

8 Out-of-Fit Split and Leakage Controls

Eligible source sessions have at least three events and do not overlap the declared 5,000-query validation subset. Their identifiers are ordered by a stable BLAKE2 fraction. At most 5,000 (10% when smaller) are held out:

- the first 1,000 become profile-lock leave-last-out queries;
- the remaining 4,000 become gate-calibration leave-last-out queries;
- all 5,000 source sessions are entirely absent from the inner CEARF index and PASGR training fit.

Consequently, the target event, its incoming transition, and every earlier event from the same source are absent from the experts that predict it. The profile/gate fingerprints are SHA-256 hashes computed before declared-validation/test prediction.

On Diginetica, declared-validation target events and incoming transitions are removed from tuning sessions; on Amazon those targets are already outside the training histories. The complete training split is restored only for the single final test expert fit. Declared-validation labels are allowed only for reporting the frozen system, never as arguments to either beta fitter. We separately record candidate-pool size, declared-subset size, and zero source overlap between the declared subset and OOF calibration.

Table 8. Fixed split identities. SHA-256 fingerprints are shown as stable abbreviated identity checks.

Domain	Eligible	Held	Profile hash	Gate hash
Video Games	89,762	5,000	623fdf1e...	3df29839...
Baby Products	145,777	5,000	b0fb3c22...	3556eb83...
Diginetica	128,724	5,000	c0f021cd...	13a5b5af...

9 Complete Allocation Results

Table 9. Allocation ablation on full-catalogue test queries. Values are mean $\pm$ SD over matched seeds; $U = .5R@6 + .5R@20$. No row uses validation to fit or select β .

Domain	Allocation	R@6	R@20	nD@20	U
Video	Memory endpoint	.06932 (.00000)	.11901 (.00000)	.06009 (.00000)	.09417 (.00000)
	Neural endpoint	.06158 (.00019)	.12571 (.00007)	.05611 (.00007)	.09364 (.00012)
	Fixed .5	.07961 (.00022)	.14610 (.00018)	.06937 (.00025)	.11285 (.00009)
	OOF global	.07799 (.00036)	.14719 (.00033)	.06886 (.00011)	.11259 (.00027)
	OOF short/long	.07830 (.00049)	.14726 (.00014)	.06900 (.00015)	.11278 (.00023)
	Dynamic β_q	.07827 (.00035)	.14735 (.00020)	.06904 (.00015)	.11281 (.00021)
Baby	Memory endpoint	.02288 (.00000)	.03879 (.00000)	.02017 (.00000)	.03084 (.00000)
	Neural endpoint	.02337 (.00060)	.04873 (.00050)	.02164 (.00040)	.03605 (.00049)
	Fixed .5	.03041 (.00038)	.05555 (.00038)	.02666 (.00017)	.04298 (.00031)
	OOF global	.03011 (.00031)	.05671 (.00042)	.02690 (.00015)	.04341 (.00031)
	OOF short/long	.03038 (.00047)	.05655 (.00053)	.02690 (.00019)	.04346 (.00040)
	Dynamic β_q	.03017 (.00037)	.05669 (.00037)	.02691 (.00016)	.04343 (.00032)
Diginetica	Memory endpoint	.28941 (.00000)	.49428 (.00000)	.24417 (.00000)	.39185 (.00000)
	Neural endpoint	.26399 (.00212)	.44852 (.00302)	.22840 (.00124)	.35626 (.00249)
	Fixed .5	.30866 (.00050)	.51430 (.00135)	.25658 (.00070)	.41148 (.00087)
	OOF global	.30958 (.00105)	.51702 (.00142)	.25786 (.00056)	.41330 (.00103)
	OOF short/long	.30926 (.00115)	.51692 (.00159)	.25775 (.00059)	.41309 (.00117)
	Dynamic β_q	.30964 (.00135)	.51701 (.00142)	.25792 (.00052)	.41332 (.00115)

Table 10. Dynamic β_q minus training-only global β ; paired query-level 95% bootstrap intervals after averaging matched-seed outcomes.

Domain		$\Delta R@6$	$\Delta R@20$	$\Delta nD@20$	ΔU			
Video	+ .00028	[+ .00011, + .00044]	+ .00016	[- .00003, + .00036]	+ .00018	[+ .00012, + .00024]	+ .00022	[+ .00009, + .00035]
Baby	+ .00005	[- .00001, + .00011]	- .00002	[- .00009, + .00004]	+ .00000	[- .00002, + .00002]	+ .00002	[- .00003, + .00006]
Diginetica	+ .00005	[- .00018, + .00030]	- .00002	[- .00024, + .00020]	+ .00006	[- .00001, + .00013]	+ .00002	[- .00015, + .00018]

Table 11. Complete allocation metrics on full-catalogue test queries. Values are mean (SD) over matched seeds.

Domain	Allocation	R@6	nD@6	R@10	nD@10	R@20	nD@20	U
Video	Fixed .5	.07961 (.00022)	.05110 (.00022)	.10505 (.00034)	.05901 (.00019)	.14610 (.00018)	.06937 (.00025)	.11285 (.00009)
	OOF global	.07799 (.00036)	.04992 (.00020)	.10347 (.00024)	.05785 (.00024)	.14719 (.00033)	.06886 (.00011)	.11259 (.00027)
	OOF short/long	.07830 (.00049)	.05012 (.00027)	.10403 (.00031)	.05811 (.00025)	.14726 (.00014)	.06900 (.00015)	.11278 (.00023)
	Dynamic β_q	.07827 (.00035)	.05012 (.00024)	.10388 (.00018)	.05809 (.00024)	.14735 (.00020)	.06904 (.00015)	.11281 (.00021)
Baby	Fixed .5	.03041 (.00038)	.01978 (.00023)	.03959 (.00028)	.02265 (.00018)	.05555 (.00038)	.02666 (.00017)	.04298 (.00031)
	OOF global	.03011 (.00031)	.01963 (.00020)	.03992 (.00029)	.02268 (.00020)	.05671 (.00042)	.02690 (.00015)	.04341 (.00031)
	OOF short/long	.03038 (.00047)	.01976 (.00025)	.03995 (.00022)	.02273 (.00017)	.05655 (.00053)	.02690 (.00019)	.04346 (.00040)
	Dynamic β_q	.03017 (.00037)	.01966 (.00022)	.03997 (.00031)	.02270 (.00020)	.05669 (.00037)	.02691 (.00016)	.04343 (.00032)
Diginetica	Fixed .5	.30866 (.00050)	.19970 (.00049)	.39255 (.00114)	.22583 (.00068)	.51430 (.00135)	.25658 (.00070)	.41148 (.00087)
	OOF global	.30958 (.00105)	.20043 (.00043)	.39470 (.00078)	.22696 (.00038)	.51702 (.00142)	.25786 (.00056)	.41330 (.00103)
	OOF short/long	.30926 (.00115)	.20028 (.00047)	.39427 (.00084)	.22677 (.00041)	.51692 (.00159)	.25775 (.00059)	.41309 (.00117)
	Dynamic β_q	.30964 (.00135)	.20051 (.00057)	.39463 (.00081)	.22700 (.00039)	.51701 (.00142)	.25792 (.00052)	.41332 (.00115)

The first table includes the two endpoints and every principal allocation control; the second gives all declared cutoffs for the non-endpoint allocation policies. Complete per-cutoff results are reported in Tables 9–11.

9.1 Per-query rescues and dynamic range

Table 12. Dynamic behavior on full test sets, mean (SD) over matched seeds. A rescue is an R@20 miss by memory changed to a hit; damage is the reverse.

Domain	Mean β_q	SD	Rescues	Damage	Net
Video Games	.572	.013	4451 (29.9)	1766 (43.3)	2685 (19.1)
Baby Products	.541	.005	3889 (111.5)	1191 (55.9)	2698 (55.8)
Diginetica	.420	.010	3076 (55.8)	1693 (43.1)	1383 (86.6)

The large net counts are mainly a fusion effect. To isolate dynamic allocation, compare with the global policy: dynamic changes mean net rescues by 15 on Video, -3 on Baby, and -1 on Diginetica.

10 Audited External-Baseline Details

The following tables report the full-catalogue baseline results used in the main-paper comparison. A dagger marks deterministic methods: their rows for seeds 42/123/456 are identical copies for schema compatibility, not independent stochastic fits. STAN uses loader session order as its recency coordinate because original timestamps are unavailable. Baseline definitions follow V-SKNN [13], STAN [3], GRU4Rec [17], NARM [9], SR-GNN [18], and SIGMA [12]; the last is our compatible reimplementaion.

No-metadata scope. The available no-metadata results do not ablate the current training-only dynamic- β method; they belong to the earlier validation-selected short/long- β family. Only the nested Diginetica result is protocol-safe as a static diagnostic.

Table 13. Validation grids and selected neighborhood configurations.

Domain	Method	Grid and selected configuration
Video Games	V-SKNN	8 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, weighting $\in \{\text{div}, \text{quadratic}\}$; selected $k = 500$, sample = 5000, div, exclude-seen=true.
Video Games	STAN	16 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, $\lambda_{spw} \in \{1.02, 2.0\}$, $\lambda_{snh} \in \{\emptyset, 5000\}$; selected $k = 500$, sample = 5000, $\lambda_{spw} = 2.0$, $\lambda_{snh} = \emptyset$, exclude-seen=true.
Baby Products	V-SKNN	8 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, weighting $\in \{\text{div}, \text{quadratic}\}$; selected $k = 500$, sample = 5000, div, exclude-seen=true.
Baby Products	STAN	16 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, $\lambda_{spw} \in \{1.02, 2.0\}$, $\lambda_{snh} \in \{\emptyset, 5000\}$; selected $k = 500$, sample = 5000, $\lambda_{spw} = 2.0$, $\lambda_{snh} = \emptyset$, exclude-seen=true.
Diginetica	V-SKNN	8 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, weighting $\in \{\text{div}, \text{quadratic}\}$; selected $k = 100$, sample = 5000, quadratic, exclude-seen=false.
Diginetica	STAN	16 cells: $k \in \{100, 500\}$, sample $\in \{1000, 5000\}$, $\lambda_{spw} \in \{1.02, 2.0\}$, $\lambda_{snh} \in \{\emptyset, 5000\}$; selected $k = 500$, sample = 5000, $\lambda_{spw} = 1.02$, $\lambda_{snh} = \emptyset$, exclude-seen=false.

Selection maximizes $0.5R@6 + 0.5R@20$ on 5,000 declared validation queries. Score weighting is div; $\lambda_{inh} = 2.05$ is fixed.

Table 14. Compact full-catalog baseline metrics on Video Games. Neural entries are mean $\pm$ sample SD over seeds 42/123/456. R@10 and earlier-cutoff nDCG are omitted for compactness.

Method	R@6	R@20	nDCG@20
Transition-only [†]	0.01712	0.04223	0.01733
V-SKNN [†]	0.06713	0.11936	0.05852
STAN [†]	0.06841	0.12115	0.05915
GRU4Rec	0.05238 $\pm$ 0.00113	0.10368 $\pm$ 0.00089	0.04733 $\pm$ 0.00088
NARM	0.06956 $\pm$ 0.00069	0.13705 $\pm$ 0.00081	0.06316 $\pm$ 0.00037
SR-GNN	0.06154 $\pm$ 0.00037	0.12267 $\pm$ 0.00060	0.05558 $\pm$ 0.00031
SIGMA-compatible	0.04367 $\pm$ 0.00104	0.08644 $\pm$ 0.00215	0.03970 $\pm$ 0.00077

Table 15. Compact full-catalog baseline metrics on Baby Products. Neural entries are mean $\pm$ sample SD over seeds 42/123/456. R@10 and earlier-cutoff nDCG are omitted for compactness.

Method	R@6	R@20	nDCG@20
Transition-only [†]	0.00299	0.00735	0.00311
V-SKNN [†]	0.02604	0.05038	0.02388
STAN [†]	0.02721	0.04944	0.02425
GRU4Rec	0.01939 $\pm$ 0.00038	0.04501 $\pm$ 0.00056	0.01882 $\pm$ 0.00019
NARM	0.01195 $\pm$ 0.00060	0.02990 $\pm$ 0.00018	0.01117 $\pm$ 0.00006
SR-GNN	0.02332 $\pm$ 0.00038	0.05208 $\pm$ 0.00127	0.02230 $\pm$ 0.00032
SIGMA-compatible	0.00856 $\pm$ 0.00096	0.02473 $\pm$ 0.00059	0.00897 $\pm$ 0.00039

Table 16. Compact full-catalog baseline metrics on Diginetica. Neural entries are mean $\pm$ sample SD over seeds 42/123/456. R@10 and earlier-cutoff nDCG are omitted for compactness.

Method	R@6	R@20	nDCG@20
Transition-only [†]	0.18392	0.40102	0.16687
V-SKNN [†]	0.30481	0.50506	0.25184
STAN [†]	0.31785	0.51502	0.25952
GRU4Rec	0.23008 $\pm$ 0.00065	0.41785 $\pm$ 0.00108	0.19557 $\pm$ 0.00030
NARM	0.31341 $\pm$ 0.00135	0.53406 $\pm$ 0.00052	0.26321 $\pm$ 0.00037
SR-GNN	0.27638 $\pm$ 0.00256	0.49267 $\pm$ 0.00468	0.23599 $\pm$ 0.00237
SIGMA-compatible	0.20528 $\pm$ 0.00589	0.37219 $\pm$ 0.01094	0.17374 $\pm$ 0.00489

Table 17. Validation-selected epochs for ID-only neural baselines.

Domain	GRU4Rec	NARM	SR-GNN	SIGMA-compatible
Video Games	12/12/12	12/12/12	5/5/5	6/6/6
Baby Products	9/8/10	1/1/3	5/5/4	1/2/1
Diginetica	12/12/12	11/12/11	4/4/5	7/9/10

Shared fixed settings: 64 dimensions, 50-item maximum context, full-catalog cross-entropy, AdamW, weight decay 10^{-5} , gradient clipping at 5, batch 512 (SR-GNN: 128), learning rate 10^{-3} (SIGMA-compatible: 5×10^{-4}). Only epoch is validation-selected; these are our local reimplementations, and SIGMA-compatible is not official SIGMA.

Table 18. Resolved semantic-teacher provenance. Reported SHA-256 prefixes identify the cached matrices used.

Domain	Resolved teacher	Shape	SHA-256
Video Games	E5-small cache	25613×384	a22a31407461
Baby Products	TF-IDF/SVD cache	36014×128	3191943cc015
Diginetica	TF-IDF/SVD product-name cache	43098×128	e31c0d501b42

Table 19. Matched-teacher NARM final-cutoff metrics on Amazon domains. Values are mean $\pm$ sample SD over seeds 42/123/456; no matched-teacher Diginetica run is retained.

Domain	R@20	nDCG@20
Video Games	0.15387 $\pm$ 0.00051	0.07112 $\pm$ 0.00029
Baby Products	0.06628 $\pm$ 0.00014	0.02914 $\pm$ 0.00010

Only the final cutoff is shown because the matched-teacher claim concerns R@20 and nDCG@20. The earlier fairness record mislabels the Video Games teacher: resolving the loaded matrix gives the higher-priority E5-small cache before TF-IDF/SVD. The Video cache has no encoder sidecar; the reported SHA-256 prefix identifies the matrix used. Exact matched-teacher NARM R@20 means are .153869695 and .066278898.

10.1 Neural-expert sensitivity

This completed 3-domain, 3-seed control replaces PASGR by ID-only NARM while holding CEARF memory, OOF split, continuous objective, bounded gate, RRF $k = 20$, and full-catalogue test protocol fixed. It is not metadata-matched on Amazon because PASGR uses semantic initialization; its role is to show that the allocator can absorb a stronger ID-only neural expert. The swap closes the Diginetica external-baseline gap (dynamic R@20 .53939 versus NARM .53406) but reduces Baby relative to the PASGR primary, consistent with Baby’s meta-data benefit.

Table 20. ID-only NARM expert swap. Values are test R@20 mean (SD) over three completed seeds. The CEARF memory, OOF split, continuous objective, bounded gate, and $k = 20$ are held fixed; the test expert is freshly refit on complete training sessions.

Domain	Neural expert	Neural-only	OOF global	Dynamic β_q
Video	NARM (ID-only)	.13763 (.00088)	.14819 (.00041)	.14854 (.00034)
Baby	NARM (ID-only)	.03880 (.00785)	.04994 (.00542)	.05001 (.00534)
Diginetica	NARM (ID-only)	.53178 (.00025)	.53910 (.00084)	.53939 (.00061)

11 Gate-Capacity Ablation

The implementation computes 16 target-free diagnostics for ablation:

Table 21. Target-free diagnostics available to gate-capacity ablations. The primary gate uses only features 1–3.

ID	Feature	ID	Feature
1	log context length	9	memory top-1 in neural top-5
2	log last-item training frequency	10	neural top-1 in memory top-5
3	last-item tail indicator	11	memory top-1 component support
4	maximum memory agreement at 5	12	neural top-1 component support
5	memory/neural top-1 equality	13	mean memory Jaccard at 10
6	memory–neural Jaccard at 5	14	neural novelty outside memory top-120
7	memory–neural Jaccard at 20	15	log transition branching factor
8	reciprocal-rank overlap at 20	16	dominant transition share

Table 22. Gate-capacity ablation, test U mean (SD) over matched seeds. “Primary” is features 1–3, linear, bounded by .10.

Gate	Video	Baby	Diginetica
Primary context linear	.11281 (.00021)	.04343 (.00032)	.41332 (.00115)
Context MLP (16 hidden)	.11303 (.00026)	.04345 (.00030)	.41331 (.00108)
All-feature linear	.11295 (.00032)	.04349 (.00041)	.41345 (.00094)
All-feature MLP	.11310 (.00044)	.04348 (.00039)	.41321 (.00105)
MLP without cross-expert features	.11297 (.00034)	.04345 (.00035)	.41304 (.00093)
MLP without memory-certainty features	.11309 (.00026)	.04347 (.00041)	.41357 (.00079)

No domain-specific post-hoc architecture is selected. The all-feature linear gate has a slightly higher mean in each domain but loses to the primary on one matched seed in each. On Diginetica, three of four MLP variants are lower than the primary and the remaining no-memory-certainty MLP is higher by only .00025 U . The same three-feature linear form and hyperparameters therefore remain primary across domains and seeds.

12 Fusion Sensitivity

Table 23. Dynamic test U mean (SD) after refitting the continuous global prior and gate on OOF training ranks for each RRF constant.

Constant	Video	Baby	Diginetica
$k = 10$	.11219 (.00018)	.04261 (.00034)	.41271 (.00081)
$k = 20$	.11281 (.00021)	.04343 (.00032)	.41332 (.00115)
$k = 60$	.11163 (.00026)	.04354 (.00027)	.41227 (.00074)

This is a sensitivity test, not a validation search: primary $k = 20$ is held fixed across domains and seeds rather than reselected from this table. Raw reciprocal ranks are never relabelled as raw expert scores. A normalized-score operator control is reported only if frozen checkpoint inference exactly reconstructs the persisted top-120 expert IDs. CombSUM min-max normalizes each expert per query over the union of those persisted top-120 candidates (absent candidates receive zero). It is evaluated under frozen β_q and parameter-free $\beta = .5$; neither allocation is reoptimized for CombSUM. This is not a full-catalogue score-normalization claim.

With the frozen dynamic allocation, weighted RRF improves R@20 over normalized CombSUM by .00133, .00134, and .00272 on Video, Baby, and Diginetica, respectively. The equal-allocation Diginetica result reverses in favor of CombSUM. Hence the control supports rank-space RRF for the proposed allocator, but it also shows an operator-allocation interaction; it does not establish unconditional RRF superiority.

Table 24. Fusion-operator perturbation under the frozen dynamic β_q and parameter-free $\beta = .5$. Values are mean (SD) over matched seeds. Intervals are paired query-level 95% bootstrap intervals for normalized CombSUM minus weighted RRF.

Domain	Allocation	RRF R@20	CombSUM R@20	$\Delta R@20$ [95% CI]
Video	Dynamic β_q	.14735 (.00020)	.14602 (.00058)	-.00133 [-.00175, -.00090]
Video	$\beta = .5$	.14610 (.00018)	.14378 (.00042)	-.00232 [-.00275, -.00190]
Baby	Dynamic β_q	.05669 (.00037)	.05534 (.00052)	-.00134 [-.00159, -.00110]
Baby	$\beta = .5$	.05555 (.00038)	.05410 (.00038)	-.00146 [-.00169, -.00122]
Diginetica	Dynamic β_q	.51701 (.00142)	.51429 (.00018)	-.00272 [-.00367, -.00176]
Diginetica	$\beta = .5$	.51430 (.00135)	.51625 (.00068)	+.00194 [+ .00095, +.00296]

The table reports the R@20 control used for the fusion-operator claim; no claim depends on omitted cutoffs.

13 Allocation Controls

Table 25: Training-only allocation controls across all domains. Learned policies use frozen OOF training ranks; reassignment is a fixed target-free control, and no validation or test labels enter fitting or selection.

Domain	Policy	R@6	R@20	nDCG@20	U
Video Games	OOF global	0.07799	0.14719	0.06886	0.11259
Video Games	Head/tail scalars	0.07821	0.14730	0.06896	0.11275
Video Games	Short/long $\times$ head/tail	0.07827	0.14728	0.06901	0.11278
Video Games	Dynamic $\Delta = .05$	0.07820	0.14729	0.06900	0.11274
Video Games	Dynamic $\Delta = .10$ (primary)	0.07827	0.14735	0.06904	0.11281
Video Games	Primary β_q reassigned	0.07817	0.14725	0.06890	0.11271
Video Games	Dynamic $\Delta = .20$	0.07840	0.14744	0.06912	0.11292
Video Games	Length only	0.07817	0.14720	0.06894	0.11269
Video Games	Frequency only	0.07819	0.14728	0.06896	0.11274
Video Games	Tail only	0.07816	0.14731	0.06895	0.11274
Video Games	Drop length	0.07819	0.14733	0.06898	0.11276
Video Games	Drop frequency	0.07824	0.14728	0.06901	0.11276
Video Games	Drop tail	0.07825	0.14734	0.06904	0.11279
Video Games	No admission cost	0.07828	0.14734	0.06904	0.11281
Video Games	No prior penalty	0.07827	0.14735	0.06904	0.11281
Baby Products	OOF global	0.03011	0.05671	0.02690	0.04341
Baby Products	Head/tail scalars	0.03018	0.05661	0.02688	0.04339
Baby Products	Short/long $\times$ head/tail	0.03018	0.05643	0.02684	0.04331
Baby Products	Dynamic $\Delta = .05$	0.03014	0.05672	0.02691	0.04343
Baby Products	Dynamic $\Delta = .10$ (primary)	0.03017	0.05669	0.02691	0.04343
Baby Products	Primary β_q reassigned	0.03018	0.05671	0.02691	0.04345
Baby Products	Dynamic $\Delta = .20$	0.03016	0.05670	0.02690	0.04343
Baby Products	Length only	0.03015	0.05670	0.02690	0.04343
Baby Products	Frequency only	0.03015	0.05666	0.02690	0.04340
Baby Products	Tail only	0.03017	0.05664	0.02689	0.04340
Baby Products	Drop length	0.03016	0.05667	0.02690	0.04341
Baby Products	Drop frequency	0.03017	0.05668	0.02690	0.04342

Table 25 – continued

Domain	Policy	R@6	R@20	nDCG@20	U
Baby Products	Drop tail	0.03016	0.05671	0.02691	0.04343
Baby Products	No admission cost	0.03017	0.05669	0.02691	0.04343
Baby Products	No prior penalty	0.03017	0.05669	0.02691	0.04343
Diginetica	OOF global	0.30958	0.51702	0.25786	0.41330
Diginetica	Head/tail scalars	0.30984	0.51707	0.25798	0.41346
Diginetica	Short/long $\times$ head/tail	0.30965	0.51695	0.25792	0.41330
Diginetica	Dynamic $\Delta = .05$	0.30961	0.51704	0.25791	0.41333
Diginetica	Dynamic $\Delta = .10$ (primary)	0.30964	0.51701	0.25792	0.41332
Diginetica	Primary β_q reassigned	0.30960	0.51685	0.25784	0.41322
Diginetica	Dynamic $\Delta = .20$	0.30956	0.51702	0.25791	0.41329
Diginetica	Length only	0.30947	0.51692	0.25781	0.41320
Diginetica	Frequency only	0.30961	0.51706	0.25791	0.41334
Diginetica	Tail only	0.30985	0.51709	0.25794	0.41347
Diginetica	Drop length	0.30974	0.51710	0.25795	0.41342
Diginetica	Drop frequency	0.30966	0.51697	0.25789	0.41331
Diginetica	Drop tail	0.30956	0.51698	0.25786	0.41327
Diginetica	No admission cost	0.30959	0.51701	0.25791	0.41330
Diginetica	No prior penalty	0.30964	0.51701	0.25792	0.41332

R@10, nDCG@6, and nDCG@10 are omitted because no allocation-control claim depends on them.

Table 26. Learned assignment versus fixed reassignment across all domains. Entries are primary minus reassigned after matched-seed averaging; paired query-level 95% bootstrap intervals are unadjusted across metrics.

Domain	Metric	Difference	95% CI
Video Games	R@20	+0.00010	[-0.00011, +0.00031]
Video Games	nDCG@6	+0.00013	[+0.00006, +0.00021]
Video Games	nDCG@10	+0.00015	[+0.00007, +0.00022]
Video Games	nDCG@20	+0.00015	[+0.00008, +0.00021]
Video Games	U	+0.00010	[-0.00003, +0.00024]
Baby Products	R@20	-0.00002	[-0.00010, +0.00006]
Baby Products	nDCG@6	0.00000	[-0.00003, +0.00002]
Baby Products	nDCG@10	+0.00001	[-0.00002, +0.00003]
Baby Products	nDCG@20	0.00000	[-0.00002, +0.00002]
Baby Products	U	-0.00002	[-0.00007, +0.00003]
Diginetica	R@20	+0.00016	[-0.00009, +0.00041]
Diginetica	nDCG@6	+0.00005	[-0.00006, +0.00015]
Diginetica	nDCG@10	+0.00013	[+0.00003, +0.00022]
Diginetica	nDCG@20	+0.00007	[0.00000, +0.00015]
Diginetica	U	+0.00010	[-0.00008, +0.00028]

Table 27. Primary linear-gate parameters on standardized OOF features, mean (sample SD) across seeds.

Domain	Length	Frequency	Tail	Bias
Video Games	-0.07662 (0.04153)	-0.09143 (0.01712)	0.01846 (0.01555)	-0.03097 (0.00398)
Baby Products	0.01041 (0.03277)	-0.00834 (0.01360)	-0.04066 (0.02517)	-0.02598 (0.00267)
Diginetica	-0.05530 (0.01823)	-0.00348 (0.02062)	0.04975 (0.05823)	-0.00651 (0.00603)

14 Statistical Procedure

For each method m , seed s , and query q , the retained per-query result contains the top-20 target rank. Recall differences are exact paired values $d_{qs} = \mathbf{1}[r_{m,qs} \leq K] - \mathbf{1}[r_{b,qs} \leq K]$. We first average d_{qs} over matched seeds, then sample query identifiers with replacement. Each sampled query carries all three seed outcomes jointly, preserving cross-seed pairing within query. Percentile endpoints from 20,000 replications form the 95% interval. A sign-flip randomization test flips each matched-seed per-query difference jointly. The evaluated HID Diginetica representation contains expanded prefix rows but not their original session identifier; the query is therefore the smallest recoverable cluster and residual within-session correlation is not modeled. These intervals describe recoverable-query sampling conditional on the three fitted seeds; sample SD across those seeds is reported separately. The procedure is not a multiple-cutoff familywise test; intervals at R@6/R@10/R@20 are descriptive matched analyses.

15 Inference Runtime and Exact Replay

Once the two top-120 expert lists are available, the gate is $O(3)$ and RRF scores at most 240 distinct items before sorting that union. The added allocation/fusion cost is therefore independent of catalogue size; full catalogue scoring remains inside the frozen experts. Features+gate and feature+gate+RRF columns in Table 28 are newly measured by exact NumPy inference. In a deterministic replay check, reconstructed beta values differ from the original inference outputs by at most 5.96×10^{-8} , and the top-20 rankings are reproduced exactly. The total is explicitly a cross-run warm component estimate: the newly measured dynamic post-processing is added to raw CEARF retrieval and PASGR inference times from the previous audited warm-expert run. It excludes every timing from the retired validation router.

Table 28. Apple M2 Pro, 32 GB, seed-42 frozen runs. Features+gate is included inside Postproc.; Expert retrieval is CEARF memory retrieval plus PASGR prediction from the prior warm-expert component run. Total = Expert retrieval + Postproc.; it is a cross-run component estimate, not a co-timed median.

Domain	Queries	Feat.+gate (μ s)	Postproc. (μ s)	Expert (ms)	Total (ms)
Video Games	94,762	1.171	85.89	3.710	3.796
Baby Products	150,777	1.184	87.57	4.031	4.119
Diginetica	60,858	.774	80.60	2.428	2.509

16 Additional Implementation Details and Limitations

All experiments use seeds 42, 123, and 456. The implementation follows the source-disjoint OOF construction, allocator optimization, paired statistical analysis, and inference procedures specified in Sections 3–15. All results required to support the paper’s claims are reported directly in the main paper and this supplementary material.

The 5,000-query cap makes calibration affordable but may undersample rare regimes. The single deterministic source holdout is out-of-fit, not full K -fold cross-fitting. The primary gate uses only last-item popularity and length context; richer uncertainty estimates may help on larger untouched domains, but the current capacity audit warns against assuming that more features are automatically safer. Finally, all main domains are next-item benchmarks; long-horizon utility and online feedback remain open.

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